RESULTS — 03-floor-pullback (pair-46 factor-base campaign, item 03)

Protocol: `PROTOCOL.md` (frozen before data), `PROTOCOL_EXT.md` (population extension E1, frozen before E1 data; also records deviation D1). Raw records: `results.json` (main run, `run.log`), `results_ext.json` (extension, `run_ext.log`; first attempt crashed in the coset arm before writing v=1549, log kept as `run_ext_attempt1_crash.log`; the fix removed a p < 1e7 guard on the divisor enumeration, and the rerun reproduced the three instances that had finished, value for value). Summary: `summary.json`. Graph: `fig_yield.svg` / `fig_yield.png`. Report: `report.pdf`.

Host: shared 4-CPU container, one core (`taskset -c 3`), Python 3 + numpy. Wall clock: main run 13 s, extension about 3 min. The budget was 40 min.

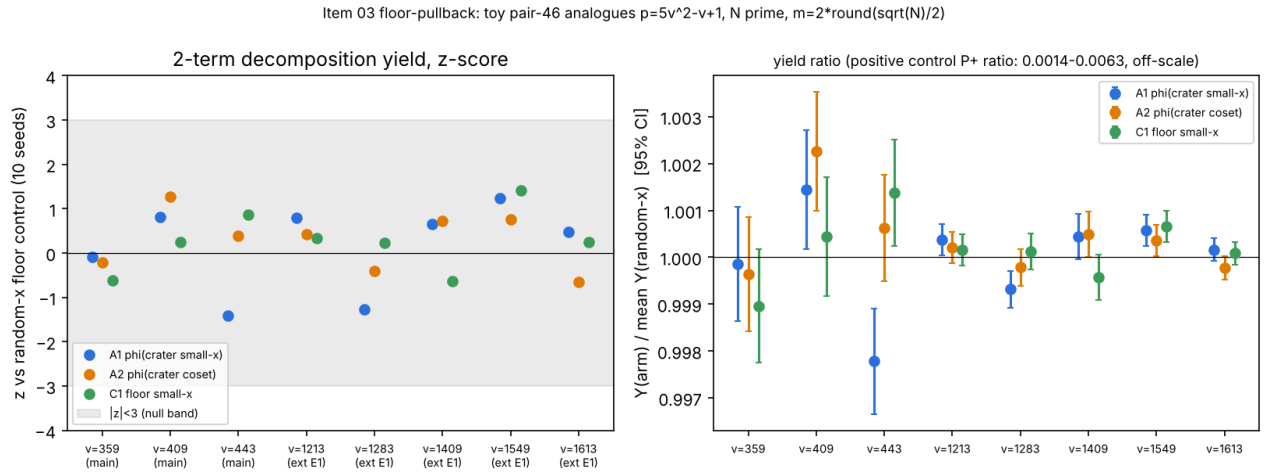


Figure 1. Left: z-score of the 2-term decomposition yield of each arm against the 10-seed random-x floor control, 8 toy instances (main + E1). Right: yield ratio to the control mean with 95% CI on that mean. Positive control off-scale (ratio <= 0.0063).

Verdict

Null (an exact obstruction, confirmed empirically). A factor base pushed from the crater through the descending v-isogeny gives the same decomposition yield on the floor as the crater base, exactly. That yield is the random-set yield: |z| <= 1.40 on all 8 toy instances for both pushed bases. Membership in the pushed base costs an extra dual-isogeny evaluation and a scalar multiplication per test, so on pair 46 this direction is strictly worse than working on the crater.

Exact statements (derivation; verified numerically on every toy)

1. N = #E(F_p) is prime and deg phi = v != N, so phi: E(F_p) -> E'(F_p) is a group isomorphism. For any S, phi(S)+phi(S) = phi(S+S), so Y_floor(phi(S)) = Y_crater(S). Every decomposition R' = phi(P1)+phi(P2) on the floor is the crater decomposition phi^-1(R') = P1+P2. The floor adds no relation that the crater lacks. Check X0: the crater yield and the floor yield, computed with two independent log tables, were bit-identical for A1 and A2 on all 8 instances.

2. Membership: Q in phi(S) iff [v^-1 mod N] * phi_hat(Q) is in S. One evaluation of the degree-v dual isogeny plus about log2 N doublings. At pair 46 that is about 6e3 field ops (sqrt-elu, figure from the task) plus about 53 doublings and 27 additions, roughly 1e3 field ops (derived). The crater small-x test is one integer comparison. In summation-polynomial form, x(phi(P)) is a rational function of degree v in x(P), so the pushed small-x set's defining relation has degree v = 3.4e7 per variable (derived). That pushes the membership degree up, the wrong way.

3. Pair 46 structure (exact for this family shape): pi + 1 = v*w and End = O_K, so the crater twist group is O_K/(v*w), which is Z/5v x Z/v. Numerical checks on the toys: on all 8, [5]P produced two independent

order-v points, so the v-torsion is fully rational. At v=359, [5v]P = O for every sampled point. This confirms the task's "derived, unverified" structure for this family shape. The pair-46 curve itself was not checked. The floor twist is cyclic (some [5v]P != O), and all v+1 v-isogenies from the crater are F_p-rational (pi acts as -1 on E[v]).

Empirical results (toy population)

Toys: p = 5v^2 - v + 1, N = 5v^2 - 2v + 4, both prime, as for pair 46. The preregistered range v in [29, 919] holds only 3 such v. E1 adds all 5 in [920, 2000]. m = 2*round(sqrt(N)/2). Metric: Y = |F+F|/N on the floor, exact.

v	p	m	mu0 (C0, 10 seeds)	sd0	z A1 pull-smallx	z A2 pull-coset	z C1 floor-smallx	P+ rati
359	644047	802	0.39365	0.00067	-0.09	-0.21	-0.61	0.0063
409	835997	914	0.39371	0.00070	+0.82	+1.28	+0.25	0.0056
443	980803	990	0.39365	0.00062	-1.40	+0.39	+0.87	0.0051
1213	7355633	2712	0.39364	0.00018	+0.80	+0.43	+0.33	0.0019
1283	8229163	2868	0.39357	0.00021	-1.27	-0.40	+0.23	0.0018
1409	9924997	3150	0.39351	0.00026	+0.66	+0.73	-0.63	0.0016
1549	11995457	3464	0.39358	0.00018	+1.23	+0.75	+1.41	0.0015
1613	13007233	3606	0.39355	0.00014	+0.47	-0.66	+0.25	0.0014

- Effect sizes: the yield ratio Y(arm)/mu0 lies in [0.9978, 1.0023] for A1 and A2 on every instance (|effect| <= 0.23%). The 95% intervals in `summary.json` cover the uncertainty in mu0 only, not the single-draw spread of the arm, so the z-score is the test. Mean z over 8 instances: A1 +0.15, A2 +0.29, C1 +0.26.

- Positive control P+ (log-space interval) has |z| >= 562 on every instance (ratio <= 0.0063). The instrument detects structured bases.

- The random-set heuristic 1 - exp(-m^2/(2N)) is 0.3932-0.3936, in line with mu0.

- Secondary metric: the fraction of pushed-base points with floor x < p/16 is 0.055-0.076 (A1) and 0.057-0.072 (A2). The C0 mean is 0.058-0.065 and uniform is 0.0625. The pushed small-x set does not stay small-x on the floor; C1 is 1.0 by construction.

- Decision rule (preregistered): transport-null is supported, with |z| < 3 for A1 and A2 everywhere and |z| > 3 for P+ everywhere.

What was cut or changed

- Pair-46-size isogeny evaluation (v = 3.4e7) was not run. The yield identity is exact, so it would add nothing. The cost there is derived, not measured.

- The preregistered main population had only 3 instances, not 6, because few v in range make both p and N prime. Extension E1 added 5 under a separate frozen protocol.

- D1: the order-v kernel points come from [5]*P, not [5v]*P as PROTOCOL.md said. The twist exponent is 5v. Arms, metric and decision are unchanged.

- Only m = 2 decompositions and one base size m ~ sqrt(N) were run. No relation-collection or linear-algebra phase was run.

Relation to prior work

This extends the isogeny-image family (EV-ICI-0bdaaa) and the small-x and coset nulls. Moving to an isogenous curve through a rational isogeny cannot change any group-theoretic yield statistic of a transported base. A floor base is only new if it is defined natively on the floor. C1 is the native floor small-x base, and it is also random-like (|z| <= 1.41).

Reproduce

```
cd research/pair46_factor_bases_20261015/03-floor-pullback
taskset -c 0 python3 run_floor_pullback.py results.json
taskset -c 0 python3 run_floor_pullback.py results_ext.json --ext
python3 analyze.py && python3 make_report.py
```